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Assignment 01 — Observability in the Wild

nopCommerce

Adding OpenTelemetry to nopCommerce

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Customer places an order

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Checkout Stage based Telemetry

Metrics:

- checkout_stage_duration_seconds
- checkout_business_failures_total
- checkout_outcomes_total

Privacy:

- allowlist instead of denylist
- checkout.variant
- store.id
- cart.items.count
- payment.method.system_name
- is_recurring
- order.total.range

Checkout Stages:

- checkout.confirm_order
- checkout.place_order
- checkout.prepare_details
- checkout.payment.process
- checkout.payment.postprocess
- checkout.order.save
- checkout.order_items.move
- checkout.inventory.adjust
- checkout.event.publish

Telemetry Mistake/Mitigation

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Intrumented:

EntityRepository.cs

EventPublisher.cs

Problem:

One operation creates multiple events and DB operations

Result:

Exponential growth of meaningless spans

Fix:

EventBus:

Span only when "OrderPlacedEvent"

Repository:

Do not span when entitie

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GenericAttribute

OrderNote

QueuedEmail

Result:

Only meanful spans are kept

Fault Injection

Level:

Implemented at the observability level, given a chance a fault might happen at a specific checkout stage

How:

ApplyInjectedDelayAsync(stage) → Task.Delay)
MaybeThrowInjectedFailure(stage) → Throws Exception

Type:

- Delay
- Failure

Purpose:

Validate the Telemetry stack

Architecture